AI ASSISTED CODING LAB ASSIGNMENT – 4

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Week: 2 **Day:** Wednesday

1. Zero-Shot Prompting – Leap Year Check

Prompt Used: Write a Python function to check whether a given year is a leap year.

Program Code:

```
def is_leap_year(year):
if (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0): return True
return False
```

Output:

Input: 2024 -> True Input: 1900 -> False Input: 2000 -> True

2. One-Shot Prompting – Centimeters to Inches Conversion

Prompt Used: Convert centimeters to inches. Example: 10 cm -> 3.94 inches

Program Code:

```
def cm_to_inches(cm): return cm / 2.54
```

Output:

Input: 10 -> 3.94

Input: 25 -> 9.84

3. Few-Shot Prompting – Name Formatting

Prompt Used: Format full name as 'Last, First' using examples.

Program Code:

```
def format_name(name):
first, last = name.split() return f"{last}, {first}
```

Output:

John Smith -> Smith, John Anita Rao -> Rao, Anita

4. Comparative Analysis – Vowel Count

Program Code:

```
def count_vowels(text): vowels = 'aeiouAEIOU' count = 0
for c in text:
if c in vowels: count += 1
return count
```

Output:

hello -> 2

education -> 5

5. Few-Shot Prompting – File Handling

Program Code:

```
def count_lines(filename):
with open(filename, 'r') as f: return len(f.readlines())
```

Output:

File with 3 lines -> 3 Empty file -> 0

Conclusion

This Lab-4 record demonstrates zero-shot, one-shot, and few-shot prompting techniques. Few-shot prompting provides better clarity, accuracy, and structured AI-generated code compared to other methods.